Data-driven output synchronization of heterogeneous leader-follower multi-agent systems

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Joint work with H. van Waarde, H.L. Trentelman, M.K. Camlibel and S. Hirche
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Chair of Information-oriented Control
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However, these data-driven methods

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Very recently, there appears a new data-driven control framework.

A new data-driven control framework

In this new framework, using directly a finite number of collected data, one can

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A new data-driven control framework

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- H. J. van Waarde, C. De Persis, M. K. Camlibel and P. Tesi, "Willems' fundamental lemma for state-space systems and its extension to multiple datasets," IEEE CSS-L, 2020.
- H. J. van Waarde, M. K. Camlibel and M. Mesbahi, "From noisy data to feedback controllers: non-conservative design via a matrix S-lemma," IEEE TAC 2021.
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Note that, however, all these results deal with single systems.

Extensions to multi-agent systems

One question then arises: is it possible to extend these results to that of multi-agent systems?

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In this talk, I will provide an affirmative answer by formulating and solving a data-driven output synchronization problem for a class of heterogeneous multi-agent systems.

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Before formulating the data-driven output synchronization problem, let us first look at the model-based case.

Model-based output synchronization problem

Leader:

$$\begin{aligned} x_r(k+1) &= Sx_r(k), \\ y_r(k) &= Rx_r(k) \end{aligned}$$

$x_r \in \mathbb{R}^r$, $y_r \in \mathbb{R}^p$, (R, S) is observable.

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Followers:

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$x_i \in \mathbb{R}^{n_i}$, $u_i \in \mathbb{R}^{m_i}$, $y_i \in \mathbb{R}^p$, $(\bar{A}_i, \bar{B}_i)$ is stabilizable, $(C_i, \bar{A}_i)$ is detectable.

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Problem: For a given communication graph, develop a distributed protocol that output synchronizes the leader-follower system.

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Question: What if the pairs $(\bar{A}_i, \bar{B}_i)$ of the followers are unknown, but only limited input and state data are available. Is it still possible to design output synchronizing protocols anymore? If so, how?

A closer look at the followers

Now, let us first take a closer look at the followers.

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Although $(\bar{A}_i, \bar{B}_i)$ are unknown, we have access to a finite set of input and state data of the followers.

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\begin{aligned}
U_{i-} &:= [u_i(0) \quad u_i(1) \quad \cdots \quad u_i(\tau-1)], \\
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Of course, the system pair $(\bar{A}_i, \bar{B}_i)$ of the ith follower explains the data (U_{i-}, X_i). However, it may not be the only system pair that explains the data!

A closer look at the followers

To see this, we partition the state data

$$
\begin{aligned}
X_{i-} &= [x_i(0) \quad x_i(1) \quad \cdots \quad x_i(\tau-1)], \\
X_{i+} &= [x_i(1) \quad x_i(2) \quad \cdots \quad x_i(\tau)], \\
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We can relate the data with

$$x_i(k+1) = \bar{A}_i x_i(k) + \bar{B}_i u_i(k)$$

via the following linear equation

$$X_{i+} = \begin{bmatrix} \bar{A}_i & \bar{B}_i \end{bmatrix} \begin{bmatrix} X_{i-} \\ U_{i-} \end{bmatrix}.$$

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Indeed, the ith follower $(\bar{A}_i, \bar{B}_i)$ is one solution of this linear equation, but it is not necessarily the unique one. There are possibly infinite many systems that explain the data, including $(\bar{A}_i, \bar{B}_i)$. And there is no way for us to distinguish $(\bar{A}_i, \bar{B}_i)$.

Now, let us keep that in mind and go back the data-driven output synchronization problem.

Data-driven output synchronization problem

Leader:

$$x_r(k+1) = Sx_r(k),$$
$$y_r(k) = Rx_r(k), \qquad \text{known}$$

Followers:

$$U_{i-} := [u_i(0) \quad u_i(1) \quad \cdots \quad u_i(\tau-1)],$$
$$X_i := [x_i(0) \quad x_i(1) \quad \cdots \quad x_i(\tau)],$$
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Aim: Find conditions under which the data (U_{i-}, X_i) are informative for the existence of distributed output synchronizing protocols. Design such protocols using (U_{i-}, X_i) (and other known information).

Data-driven output synchronization problem

Moreover, we consider the particular distributed protocol[1]

$$v_i(k+1) = Sv_i(k) + (1+d_i+g_i)^{-1}F\Big(\sum_{j=1}^{N} a_{ij}\big(v_j(k)-v_i(k)\big) + g_i\big(x_r(k)-v_i(k)\big)\Big),$$

$$u_i(k) = K_i\Big(x_i(k) - \Pi_i v_i(k)\Big) + \Gamma_i v_i(k), \quad i = 1, 2, \ldots, N \qquad (\star)$$

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We also make the following assumptions.

Assumption 1: The communication between the followers is a directed graph that contains a spanning tree; the leader is connected to at least one of the root nodes.

Assumption 2: All eigenvalues of S are simple and lie on the unit disk.

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For the followers, recall that

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\end{aligned}
$$

Note that

- we only have access to the input and state data (U_{i-}, X_i), but not the follower matrices $(\bar{A}_i, \bar{B}_i)$.
- there may have infinite many systems that explain the data

$$
X_{i+} = \begin{bmatrix} A_i & B_i \end{bmatrix} \begin{bmatrix} X_{i-} \\ U_{i-} \end{bmatrix}
$$

- it is impossible for us to identify the ith follower $(\bar{A}_i, \bar{B}_i)$ from the other systems.

Data-driven output synchronization problem

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What can we do if we still want to achieve output synchronization?

We define the set of all systems (A_i, B_i) that explain (U_{i-}, X_i) to be

$$\Sigma_i := \left\{ (A_i, B_i) \mid X_{i+} = \begin{bmatrix} A_i & B_i \end{bmatrix} \begin{bmatrix} X_{i-} \\ U_{i-} \end{bmatrix} \right\}.$$

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Clearly, $(\bar{A}_i, \bar{B}_i) \in \Sigma_i$.

Definition 1 (informativity for output synchronization)

The data (U_{i-}, X_i) are informative for output synchronization if there exists a protocol $(\star)$ that achieves output synchronization for the leader and all systems $(A_i, B_i) \in \Sigma_i$, $i = 1, 2, \ldots, N$.

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Distributed protocol:

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Assumptions on the matrix S and communication graph.

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$$v_i(k+1) = Sv_i(k) + (1+d_i+g_i)^{-1}F\Big(\sum_{j=1}^{N} a_{ij}\big(v_j(k)-v_i(k)\big) + g_i\big(x_r(k)-v_i(k)\big)\Big),$$
$$u_i(k) = K_i\Big(x_i(k) - \Pi_i v_i(k)\Big) + \Gamma_i v_i(k), \quad i = 1,2,\ldots,N \qquad (\star)$$

Assumptions on the matrix S and communication graph.

Problem: Find conditions under which the data (U_{i-}, X_i) are informative for output synchronization; and provide a design method for computing such protocols.

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Main results

Theorem 1

Let Assumptions hold. The data (U_{i-}, X_i) are informative for output synchronization if and only if, for $i = 1, 2, \ldots, N$, there exists a right-inverse $X_{i-}^{\dagger}$ of X_{i-} such that $X_{i+}X_{i-}^{\dagger}$ is stable, and, in addition, there exist matrices M_i satisfying the linear equations

$$X_{i+}M_i - X_{i-}M_iS = 0$$
$$C_iX_{i-}M_i + D_iU_{i-}M_i = R, \quad i = 1, 2, \ldots, N.$$

In this case, a protocol $(\star)$ can be found as follows: take F such that $S - \lambda_i F$ are stable[a], where λ_i are the eigenvalues of $(I_N + \mathcal{D} + G)^{-1}(L + G)$, and define

$$K_i = U_{i-}X_{i-}^{\dagger}, \quad \Pi_i = X_{i-}M_i, \quad \Gamma_i = U_{i-}M_i.$$

[a] K. Hengster-Movric, K. You, F. L. Lewis, L. Xie, "Synchronization of discrete-time multi-agent systems on graphs using Riccati design," Automatica, 2013.

Proof

For a proof of Theorem 1, please see our paper

J. Jiao, H. J. van Waarde, H. L. Trentelman, M. K. Camlibel and S. Hirche, "Data-driven output synchronization of heterogeneous leader-follower multi-agent systems," arXiv, 2021.

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There, we have also considered the case of multi-agent systems with known external disturbances.

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Simulation example

Leader:

$$x_r(k+1) = Sx_r(k),$$
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Simulation example

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$$x_r(k+1) = Sx_r(k), \qquad S = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 0 \end{bmatrix}. \qquad \text{known}$$
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$$\bar{A}_2 = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}, \quad \bar{B}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad C_2 = \begin{bmatrix} -1 & 1 \end{bmatrix}, \quad D_2 = 2,$$
$$\bar{A}_3 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \quad \bar{B}_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad C_3 = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad D_3 = 0.5.$$

Simulation example

Distributed protocol:

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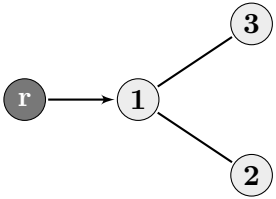
Communication graph:



Simulation example

For each follower, we collect four sets of state data and three sets of input data as follows

$$X_1 = \begin{bmatrix} 1 & 0 & 1 & 1 \\ -1 & 0 & 0 & 1 \end{bmatrix}, \quad U_{1-} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix},$$

$$X_2 = \begin{bmatrix} 1 & 0 & 3 & -1 \\ -1 & 2 & -2 & 5 \end{bmatrix}, \quad U_{2-} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix},$$

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There exists a right-inverse $X_{i-}^{\dagger}$ of X_{i-} such that $X_{i+}X_{i-}^{\dagger}$ is stable for $i = 1, 2, 3$.

Simulation example

Similarly, we compute solutions M_i to the linear equations in Theorem 1 and obtain

$$M_1 = \begin{bmatrix} 0 & 1 \\ 2 & -1 \\ -1 & 0 \end{bmatrix}, \quad M_2 = \begin{bmatrix} 0.4 & -1.4 \\ 0.4 & 0.6 \\ 0.2 & 0.8 \end{bmatrix}, \quad M_3 = \begin{bmatrix} 0.6 & -0.1 \\ -0.3 & 0.3 \\ 0.7 & -0.2 \end{bmatrix}.$$

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Next, using directly these data (U_{i-}, X_i), we compute feedback gains

$$K_1 = \begin{bmatrix} -0.4273 & -1.4516 \end{bmatrix}, \quad K_2 = \begin{bmatrix} 0.4663 & -1.4755 \end{bmatrix}, \quad K_3 = \begin{bmatrix} 0.0157 & 1.0071 \end{bmatrix}.$$

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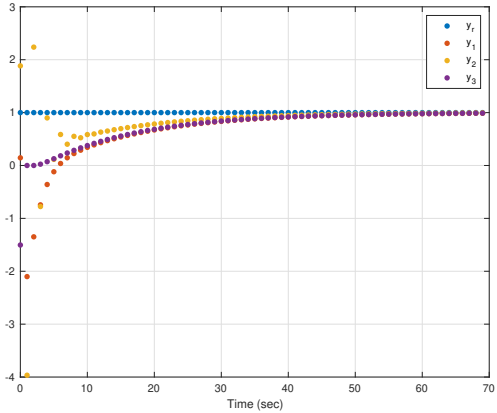
Subsequently, we also compute gain matrices F, Π_i and Γ_i as in Theorem 1.

Simulation example

The figure below plots the output trajectories y_r, y_1, y_2 and y_3. It can be seen that indeed the corresponding protocol achieves output synchronization.



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Conclusions

In this talk,

- we have formulated a data-driven output synchronization problem for heterogeneous leader-follower multi-agent systems.

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Conclusions

In this talk,

- we have formulated a data-driven output synchronization problem for heterogeneous leader-follower multi-agent systems.
- We have provided necessary and sufficient data-based conditions for output synchronization.
- We have also provided a design method for computing such output synchronizing protocols directly from data.

Future work

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There are several possibilities for future work. For example, one could consider

- the case that the external disturbance is unknown.
- the case that only input and output data are available.
- the case with time delay.
- the case with homogeneous multi-agent systems.

Related work

- J. Jiao, H. J. van Waarde, H. L. Trentelman, M. K. Camlibel and S. Hirche, "Data-driven output synchronization of heterogeneous leader-follower multi-agent systems," arXiv, 2021.
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- J. G. Rueda-Escobedo, E. Fridman, J. Schiffer, "Data-driven control for linear discrete-time delay systems," arXiv, 2020.

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